PUSHFORWARD ALONG AN INTEGRAL UNIVERSALLY INJECTIVE MORPHISM IS FULLY FAITHFUL, WITH ESSENTIAL IMAGE DESCRIBED BY THE IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the factorization is written out and both factors are handled by the named companion results, which together give both parts of the statement and the left-inverse assertion.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Factorization of an integral universally injective morphism. Same chapter, the reduction of the proposition on properties of the small pushforward to the closed immersion case, in which f is factored as a surjective morphism followed by a closed immersion. Supplies the factorization with which the statement is reduced to two known cases.
- Pushforward along a closed immersion. Same chapter, Proposition proposition-closed-immersion-pushforward: full faithfulness with essential image described by the complement of the closed subscheme and by supports. Supplies both parts of the statement for the closed immersion factor.
- Topological invariance of the etale site. Same chapter, Theorem theorem-topological-invariance. Supplies both parts of the statement for the integral, universally injective and surjective factor.

Source reference. Original proposition 04CA: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6702–6702. Navigation pointer invoked at proof line 6762; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original theorem 04DZ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6354–6354. Navigation pointer invoked at proof line 6805; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let $f : X \to Y$ be a morphism of schemes which is integral and universally injective.*

(1) *The functor*
$$f_{small,*} : Sh(X_{étale}) \longrightarrow Sh(Y_{étale})$$
is fully faithful and its essential image is those sheaves of sets $\mathcal{F}$ on $Y_{étale}$ whose restriction to $Y \setminus f(X)$ is isomorphic to $*$, and

(2) *the functor*
$$f_{small,*} : Ab(X_{étale}) \longrightarrow Ab(Y_{étale})$$
is fully faithful and its essential image is those abelian sheaves on $Y_{étale}$ whose support is contained in $f(X)$.

In both cases f_{small}^{-1} is a left inverse to the functor $f_{small,*}$.

Proof. We may factor f as
$$X \xrightarrow{h} Z \xrightarrow{i} Y$$
where h is integral, universally injective and surjective and $i : Z \to Y$ is a closed immersion. Apply Proposition 04CA to i and apply Theorem 04DZ to h. □